

ASSESSMENT GUIDE / MARKING SCHEME

Specialist Mathematics Unit 2 - Number and Proof

Award marks for mathematically equivalent reasoning. Where a proof is requested, unsupported answers do not earn the proof marks. Consequential working may be credited where appropriate.

Question 1 - 4 marks

a. Identifies $-\sqrt{3}$ as an irrational algebraic number (hence not transcendental).	1
Provides a valid classification/justification, e.g. it is a root of $x^2 - 3 = 0$ but is not rational. Do not require classification of every other listed number.	1
b. Forms the required expression: $n(n+1)+(n+1)$.	1
Factorises correctly to $(n+1)^2$ and concludes this is a perfect square for integer n .	1

Question 2 - 4 marks

a. "If a student skips the house event, then the student is not hypnotized."	1
b. States the contrapositive to be proved: if n is even, then n^2-4n+7 is odd.	1
Lets $n=2k$, $k \in \mathbb{Z}$, and substitutes to obtain $4k^2-8k+7 = 2(2k^2-4k+3)+1$.	1
Concludes the expression is odd, so the contrapositive and hence the original implication are true.	1

Question 3 - 4 marks

Assumes for contradiction that $\sqrt{5}$ is rational and writes $\sqrt{5} = a/b$ in lowest terms, with integers a, b and $b \neq 0$.	1
Squares to obtain $a^2 = 5b^2$, and deduces 5 divides a .	1
Writes $a = 5k$, substitutes, and deduces 5 divides b .	1
Contradicts lowest terms and concludes $\sqrt{5}$ is irrational.	1

Question 4 - 5 marks

Correct base case, e.g. $n=1$: $1(1!) = 1 = 2! - 1$.	1
States the induction hypothesis for $n=k$: $\sum_{r=1}^k r(r!) = (k+1)! - 1$.	1
For $n=k+1$, writes $\sum_{r=1}^{k+1} r(r!) = [(k+1)! - 1] + (k+1)(k+1)!$.	1
Factorises/simplifies to $(k+1)![1 + (k+1)] - 1 = (k+2)! - 1$.	1
States the induction conclusion for all positive integers n .	1

Question 5 - 6 marks

Correct base case $n=1$: $M = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$, matching the claimed form.	1
States the induction hypothesis $M^k = \begin{bmatrix} 2^k & 2^k-1 \\ 0 & 1 \end{bmatrix}$.	1
Writes $M^{k+1} = M^k M$.	1
Performs the matrix multiplication correctly.	1
Obtains $M^{k+1} = \begin{bmatrix} 2^{k+1} & 2^{k+1}-1 \\ 0 & 1 \end{bmatrix}$.	1
States the induction conclusion for all positive integers n .	1

Question 6 - 5 marks

Checks the base case $n=5$: $25 < 32$.	1
Assumes $k^2 < 2^k$ for some integer $k \geq 5$.	1
Establishes $(k+1)^2 < 2k^2$ for $k \geq 5$, e.g. $k^2 - 2k - 1 > 0$.	1
Uses the hypothesis: $(k+1)^2 < 2k^2 < 2 \cdot 2^k = 2^{k+1}$.	1
States the induction conclusion for all integers $n \geq 5$.	1

Question 7 - 3 marks

a. Gives a valid counterexample, e.g. $n=40$ gives $40^2+40+41=1681=41^2$, which is composite. **1**

b. Correct base case $n=2$: $1-1/4=3/4=(2+1)/(2\cdot 2)$. **1**

Induction step: assumes $P(k)$, multiplies by $1-1/(k+1)^2$, simplifies to $(k+2)/(2(k+1))$, and states the induction conclusion. **1**

TOTAL: 31 MARKS